

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter-2023 Course: B. Tech. Branch : Computer & Allied Engineering Semester :IV Subject Code & Name: BTCOC401 Design & Analysis of Algorithms Max Marks: 60 Date:16-01-24 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	What is Algorithm? Explain criteria of Algorithms.	Remember	6
B)	What is Asymptotic Notations? Explain any three Asymptotic Notations.	Remember	6
C)	Define Max and Min Heap and write algorithm to insert into a heap.	Analysis	6
Q.2	Solve Any Two of the following.		12
A)	Write algorithm for Binary Search and calculate its time complexity.	Application	6
B)	Explain Quick Sort algorithm with its performance analysis.	Analysis	6
C)	Explain Strassen's Matrix Multiplication.	Remember	6
Q. 3	Solve Any Two of the following.		12
A)	Draw a state space tree for finding Four Queens problems solution.	Application	6
B)	Describe Graph Coloring Problem with suitable example	Analysis	6
C)	Explain Branch and Bound with suitable example.	Remember	6
Q.4	Solve Any Two of the following.		12
A)	Find an optimal solution to the knapsack instance $n = 7$, $m = 15$, $(P_1, P_2, P_3, P_4, P_5, P_6, P_7) = (10, 5, 15, 7, 6, 18, 3)$, and $(w_1, w_2, w_3, w_4, w_5, w_6, w_7) = (2, 3, 5, 7, 1, 4, 1)$.	Application	6
B)	What is Minimum Cost Spanning Tree? Explain with suitable example.	Remember	6
C)	Explain Job Sequencing with Deadlines.	Remember	6
Q. 5	Solve Any Two of the following.		12
A)	Analyze Floyd Warshals algorithm for Dynamic Programming.	Analysis	6
B)	Explain complexity class P and complexity class NP.	Remember	6
C)	Differentiate between Dynamic Programming and Greedy Algorithm.	Analysis	6